1. Clinical Question & Core Hypothesis

Clinical Question

NMIBC tumors often show abundant intratumoral lymphocytes yet vary widely in progression; why does high TIL density sometimes fail to translate into durable tumor control? 2) Tier 1 paths used: I built on PSMD2 (proteostasis/proteasome regulation of p53 and antigen-processing) and BRG1 (SWI/SNF chromatin remodeling that can repress antigen-presentation and chemokine programs). These two provide complementary mechanisms that can jointly disable T-cell–mediated control despite TIL presence. 3) Tier 2 evidence supporting the claim: metabolic enzymes (ADH4) and translational/ ncRNA programs can alter redox/antigenicity and cytokine/chemokine outputs; CONUT score captures host nutritional–immune competence that modulates whether intratumoral lymphocytes are effective; pluripotency/PIK3CA–linked stemness can increase immune evasion. 4) Why PFI: progression-free interval directly reflects NMIBC progression/recurrence risk (the clinical problem implied by the Ontologist) and is sensitive to incomplete immune control, making it the most appropriate single endpoint. 5) Deliberate exclusions: I excluded exploratory Tier 3 signals (STMY3, chr10 q11.2 locus, AL357033.4 lncRNA) from the core hypothesis because they lack mechanistic specificity; I also limited core anchors to two genes (PSMD2, BRG1) to keep the hypothesis sharply testable, moving GARS1 and other Tier2 modifiers into secondary_modifiers for follow-up analyses?

Working Hypothesis

Among NMIBC tumors with high TIL density, progression risk is elevated when concurrent PSMD2-driven proteasomal activity and BRG1-mediated chromatin remodeling suppress effective antigen processing/presentation and dampen chemokine/MHC signals necessary for cytotoxic T cell function. This core axis creates an immune-evading niche in which TILs are present but functionally impaired, leading to shorter progression-free interval (PFI). The effect is modulated by Tier2 modifiers: elevated GARS1 (translational/proteostatic flux), ADH4-driven metabolic/redox states, ncRNA programs that regulate immune and antigen-presentation genes, CONUT-detected nutritional/immune competence, and PIK3CA-driven PI3K signaling that promotes immune evasion. Collectively, tumors with high PSMD2 and BRG1 activity and adverse Tier2 modifier profiles will exhibit reduced MHC class I expression, diminished TAP1/LMP2 function, lower CXCL9/CXCL10 chemokine outputs, and decreased T cell cytotoxicity, translating into poorer PFI despite abundant TILs. In contrast, tumors with lower PSMD2/BRG1 activity or favorable Tier2 modifiers should show restored antigen presentation, enhanced T cell–mediated killing, and longer PFIs. This leads to testable predictions about molecular readouts, immune function assays, and clinical outcomes that can be evaluated in NMIBC cohorts and mechanistic models.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	PSMD2, BRG1
Primary endpoint	PFI

2. Why This Gene — Not Others

Primary hypothesis gene(s)

PSMD2: PSMD2 (proteostasis/proteasome regulation of p53 and antigen-processing) and BRG1 (SWI/SNF chromatin remodeling that can repress antigen-presentation and chemokine programs).

BRG1: BRG1) to keep the hypothesis sharply testable, moving GARS1 and other Tier2 modifiers into secondary_modifiers for follow-up analyses.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	GARS1	Supporting modifier from knowledge graph
Secondary	ADH4	Supporting modifier from knowledge graph
Secondary	ncRNAs	Supporting modifier from knowledge graph
Secondary	CONUT score	Supporting modifier from knowledge graph
Secondary	PIK3CA	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In functional experiments, restoring or enhancing the anchor gene's signaling will improve immune-mediated tumor control in imaging-feature-positive models.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Gene expression does not equal protein activity; phospho-protein or functional assays are needed for confirmation.

Alternative Directions

- GARS1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- ADH4: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.